Project: Food Delivery Service

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Entity Identification + Attribute Definition + PK/FK Identification (Table View, Written View Is Below)

Customers

CustomerID (PK)
FirstName
LastName
Email
PhoneNumber
PasswordHash

CustomerAddresses

AddressID (PK)
CustomerID (FK)
Street
City
State
ZipCode

PaymentMethods

PaymentID (PK)
CustomerID (FK)
BillingAddressID (FK)
Method (Card / Digital Wallet)
Provider (Visa//Mastercard/Paypal/etc.)

Token (Reference to Card)
LastFour
MonthExpire
YearExpire

Restaurants

RestaurantID (PK)
Name
PhoneNumber
Email
CuisineCategory

RestaurantAddresses

RestaurantAddressID (PK)
RestaurantID (FK)
Street
City
State
ZipCode

MenuItems

ItemID (PK)
RestaurantID (FK)
Name
Description
Price
Availability

Orders

OrderID (PK)
CustomerID (FK)
RestaurantID (FK)
AddressID (FK)
PaymentID (FK)
OrderDate
Status (Preparing, Out For Delivery, Delivered)
PaymentStatus (Paid, Pending, Refunded)
SubTotal
DeliveryFee
TaxAmount
DiscountAmount
TipAmount
TotalAmount

OrderItems

OrderItemID (PK)
OrderID (FK)
ItemID (FK)
Quantity
SpecialInstructions

DeliveryPersonnel

DeliveryPersonID (PK)
Name
PhoneNumber

AvailabilityStatus
PasswordHash

Deliveries

DeliveryID (PK)
DeliveryPersonID (FK)
OrderID (FK)
AssignedTime
DeliveryTime
EstimatedTime
CurrentLocation

Written Section

Customers: CustomerID, FirstName, Last Name, Email, PhoneNumber, PasswordHash

CustomerAddresses: AddressID, CustomerID, Street, City, State, ZipCode

PaymentMethods: PaymentID, CustomerId, BillingAddressID, Method, Provider, Token, LastFour, MonthExpire, YearExpire

Restaurants: RestaurantID, Name, PhoneNumber, Email, CuisineCategory

RestaurantAddress: RestaurantAddressID, RestaurantID, Street, City, State, ZipCode

MenuItems: ItemID, RestaurantID, Name, Description, Price, Availability

Orders: OrderID, CustomerID, RestaurantID, AddressID, PaymentID, OrderDate, Status, PaymentStatus, SubTotal, DeliveryFee, TaxAmount, DiscountAmount, TipAmount, TotalAmount

OrderItems: OrderItemID, OrderID, ItemID, Quantity, SpecialInstructions

DeliveryPersonnel: DeliveryPersonID, Name, PhoneNumber, AvailabilityStatus, PasswordHash

Deliveries: DeliveryID, DeliveryPersonID, OrderID, AssignedTime, DeliveryTime, EstimatedTime, CurrentLocation

Primary/Foreign Keys

Primary:
Customers -> CustomerID
CustomerAddresses -> AddressID
PaymentMethods -> PaymentID
Restaurants -> RestaurantID
RestaurantAddress -> RestaurantAddressID
MenuItems -> ItemID
Orders -> OrderID
OrderItems -> OrderItemID
DeliveryPersonnel -> DeliveryPersonID
Deliveries -> DeliveryID

Foreign:
CustomerAddresses.CustomerID -> Customers

PaymentMethods.CustomerID -> Customers
PaymentMethods.BillingAddressID -> CustomerAddresses

RestaurantAddress.RestaurantID -> Restaurants

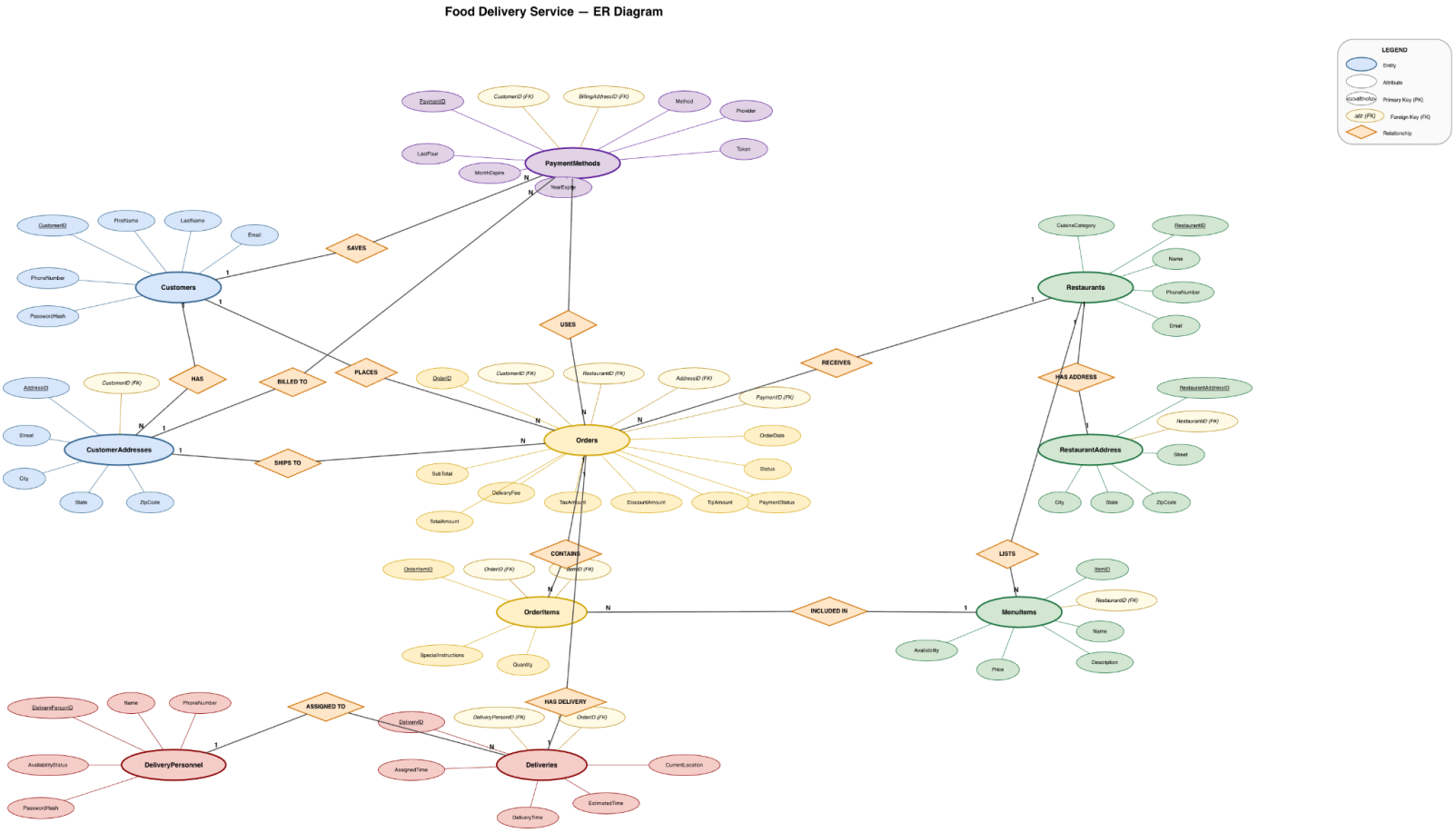
MenuItems.RestaurantID -> Restaurants

Orders.CustomerID -> Customers
Orders.RestaurantID -> Restaurants
Orders.AddressID -> CustomerAddresses
Orders.PaymentID -> PaymentMethods

OrderItems.OrderID -> Orders
OrderItems.ItemID -> MenuItems

Deliveries.DeliveryPersonID -> DeliveryPersonnel
Deliveries.OrderID -> Orders

ER Diagram



Group Tasks

Calvin Lin:

1. Documentation Creation
2. Wrote down Entities/Attributes listed in both Table View and Written View
3. Ensured Proper Primary/Foreign Keys assigned to each Entity
4. Created script for sample Restaurants, RestaurantAddresses, and MenuItems data

Anna Chapko:

Jung Chen:

1. Analysis of project requirements
2. Initial DB draft
3. Setting DB naming conventions
4. Created finalized DB script
5. Created script for sample Customers, CustomerAddresses, PaymentMethods data

Project Features:

1. Customer Management

The schema establishes a clear relationship between users and their logistics.

- Unique Profiles: The Customers table captures necessary contact details, including FirstName, LastName, Email, and PhoneNumber.
- Unique Identity: The Email field is marked as UNIQUE to prevent duplicate accounts.
- Address Management: Multiple locations (e.g., "Home" vs. "Work") can be stored in the CustomerAddresses table, linked via CustomerID.
- Secure Payments: The PaymentMethods table uses a Token system to store a reference to the card rather than raw data, while also linking to a BillingAddressID.

2. Restaurant & Menu Management

The design allows restaurants to manage their virtual storefronts dynamically.

- Restaurant Profiles: The Restaurants table stores the business name, contact info, and CuisineCategory.
- Physical Location: Business addresses are handled separately in the RestaurantAddresses table.
- Menu Items: The MenuItems table includes the essential attributes: Name, Description, Price, and Availability.
- Menu Updates: Because Price and Availability are stored at the item level, restaurants can update these as needed to reflect current stock or pricing changes.

3. Order Processing & Tracking

The core of the system handles the transition from a customer's cart to their front door.

- Order Linking: Every Order record links the CustomerID, RestaurantID, and AddressID to ensure all parties know where the food is coming from and where it is going.
- Detailed Line Items: The OrderItems table tracks the specific ItemID, Quantity, and any SpecialInstructions for the kitchen.
- Status Management: The Orders table uses an ENUM to track the lifecycle from Preparing to Out For Delivery and Delivered.
- Fulfillment & Tracking: The Deliveries table manages the assignment of DeliveryPersonnel, tracks EstimatedTime versus DeliveryTime, and provides a CurrentLocation field for real-time updates.

4. Financials & Personnel

- Charge Summaries: The Orders table provides a full financial breakdown, including SubTotal, DeliveryFee, TaxAmount, DiscountAmount, and TipAmount.
- Payment Status: The PaymentStatus column allows for tracking if an order is Paid, Pending, or Refunded.
- Driver Profiles: The DeliveryPersonnel table includes Name, PhoneNumber, and an AvailabilityStatus to ensure the system only assigns orders to active drivers.